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WRITTEN REPORT

Analysis of Augmented Krylov Subspace Methods

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1 Introduction

Saad's article [1] analyzes advancements in enhancing convergence rates within standard Krylov subspace methods such as GMRES [2], in relation to the resolution of sparse linear systems of the form $Ax = b$, by augmenting these subspaces through various methods. This augmentation aims to accelerate convergence, particularly when the spectrum of A is poorly distributed and some eigenvalues are close to zero. This is due to Krylov methods essentially approximate the solution $x = A^{-1}b$, by building polynomials that approximate the inverse of the matrix A . Hence, if A has eigenvalues close to zero, these polynomials must be high-degree to achieve a good approximation over the entire spectrum of eigenvalues, which leads to slower convergence.

The above mentioned paper presents a theoretical exploration of these augmented Krylov methods, defining them as projection methods where the projection subspace $\mathcal{K}$ combines a standard Krylov subspace, denoted as $\mathcal{K}_m$, with an additional subspace $\mathcal{W}$. This supplementary space aims to capture components of the matrix A that may be overlooked by the standard Krylov subspace dominated by large eigenvalues. The choice of $\mathcal{W}$ varies, with options including deflation techniques, where $\mathcal{W}$ represents an approximate invariant subspace typically associated with the smallest eigenvalues, and block-Krylov methods, where $\mathcal{W}$ consists of other Krylov subspaces generated with the same matrix A but different initial residuals. In both cases, residual bounds are determined, for A symmetric and non-symmetric, that indicate a convergence rate similar to what is achieved by excluding the components of the initial residual vector associated to the eigenvalues nearest to zero.

2 Augmented Krylov methods and FGMRES

Consider in what follows x_0 is the initial guess used for solving the linear system, r_0 the associated initial residual $r_0 = b - Ax_0$ and $\mathcal{K}_m(A, r_0) = \text{span}\{r_0, Ar_0, \dots, A^{m-1}r_0\}$.

Definition. *An augmented Krylov subspace method is any Krylov projection method for solving $Ax = b$ where the standard Krylov space is augmented by another subspace $\mathcal{W}$ such that the resulting solution search space is*

$$\mathcal{K} = \mathcal{K}_m + \mathcal{W}$$

Assuming we have $\mathcal{K} = \text{span}\{v_1, v_2, \dots, v_m, w_1, w_2, \dots, w_p\}$ where $\{v_1, \dots, v_m\}$ is a basis for $\mathcal{K}_m$ and $\{w_1, \dots, w_p\}$ is a basis for $\mathcal{W}$, not necessarily independent of $\mathcal{K}_m$, then a convenient implementation to obtain an orthonormal basis of $\mathcal{K}$ and subsequently minimize the system's residual is achieved through a Flexible Generalized Minimal Residual (FGMRES) method, as detailed below in Algorithm 1.

Algorithm 1: FGMRES: GMRES with variable preconditioning [3]

Step 1. Start: Choose x_0 and $m + p = \text{Krylov dim.}$ Initialize $\tilde{H}_{m+p} = 0$;

Step 2. Arnoldi iteration: $r_0 = b - Ax_0$, $\beta = \|r_0\|_2$, $v_1 = r_0/\beta$;

for $j = 1, 2, \dots, m + p$ **do**

if $j \leq m$ **then** $z_j := M_j^{-1}v_j$ **else** $z_j := M_j^{-1}w_{j-m}$;

$w := Az_j$;

for $i = 1, 2, \dots, j$ **do**

$|$ $h_{i,j} := (w, v_i)$, $w := w - h_{i,j}v_i$

end

$h_{j+1,j} = \|w\|_2$, $v_{j+1} = w/h_{j+1,j}$;

end

$Z_{m+p} := [z_1, \dots, z_{m+p}]$, $\tilde{H}_{m+p} = [h_{i,j}]_{1 \leq i \leq j+1, 1 \leq j \leq m+p}$;

Step 3. Form approx. sol.: Compute y s.t. $\min. \|\beta e_1 - \tilde{H}_{m+p}y\|_2$;

$x_m = x_0 + Z_{m+p}y$;

Step 4. Restart: If satisfied Stop, else set $x_0 \leftarrow x_m$ and goto Step 2.

The main properties of such FGMRES implementation can be summarized in:

- (i) x_m minimizes $\|b - Ax_m\|_2$ over $x_0 + \text{span}\{Z_{m+p}\}$ by satisfying the Arnoldi-like relation

$$AZ_{m+p} = V_{m+p+1}\tilde{H}_{m+p}$$

where $V_{m+p+1} = [v_1, \dots, v_{m+p+1}]$.

- (ii) If $w := Az_j = v_j$, i.e., preconditioning is exact at step j for any v_j from $1 \leq j \leq m+1$, then w depends linearly on the previous v_i (it is equal to v_j) and as a result the orthogonalization process will yield $h_{j+1,j} = 0$ (breakdown). In addition, if we assume $\tilde{H}_j$ is nonsingular, then FGMRES finds the exact solution at x_j .
- (iii) If M_j^{-1} is constant, the algorithm is equivalent to a Right-preconditioned GMRES [3].
- (iv) This method offers improved flexibility compared to a standard Right-preconditioned GMRES but adds an additional cost due to the need to store additional vectors $\{z_j\}_{j=1, \dots, m+p}$.

Thus, from property (ii) if we let z_j be any vector in $\mathcal{W}$ such that $Aw = v_j$ (invariance) with the assumption $\tilde{H}_j$ is invertible, then the affine space $x_0 + \mathcal{K}_m + \mathcal{W}$ will contain an exact solution to the linear system $Ax = b$. This, combined with the fact that poorly represented areas in the spectrum of A that slow down convergence in iterative Krylov methods such as GMRES are associated with eigenvalues close to zero, leads to the idea proposed by Saad in [1] to augment $\mathcal{K}_m$ with an approximate invariant subspace $\mathcal{W}$ associated with the small eigenvalues of A .

3 Augmenting with nearly invariant subspaces

This way, we assume the existence of an invariant subspace that is nearly close to $\mathcal{W}$ and investigate the performance of the augmented Krylov subspace algorithm in this scenario. The aim is to prove a residual bound that indicates faster convergence when the invariant subspace closely approximates $\mathcal{W}$.

3.1 General result

In order to give a measure of the concept of 'near-invariance', we need to introduce the definition of gap between two subspaces, $\Theta(\mathcal{X}, \mathcal{Y})$ [4].

Definition. For any pair of subspaces of $\mathbb{C}^n$, define

$$\delta(\mathcal{X}, \mathcal{Y}) := \max_{x \in \mathcal{X}, x \neq 0} \min_{y \in \mathcal{Y}} \frac{\|x - y\|_2}{\|x\|_2} = \|(I - P_{\mathcal{Y}})P_{\mathcal{X}}\|_2$$

where $P_{\mathcal{X}}, P_{\mathcal{Y}}$ are orth. projectors onto $\mathcal{X}, \mathcal{Y}$ resp. Then, the gap between $\mathcal{X}$ and $\mathcal{Y}$ is

$$\text{gap}(\mathcal{X}, \mathcal{Y}) \equiv \Theta(\mathcal{X}, \mathcal{Y}) := \max[\delta(\mathcal{X}, \mathcal{Y}), \delta(\mathcal{Y}, \mathcal{X})].$$

Hence, $\Theta(\mathcal{X}, \mathcal{Y})$ can be interpreted as the sine of the angle formed between the two subspaces, $\mathcal{X}$ and $\mathcal{Y}$.

A residual bound for $\mathcal{K}$ is then obtained using the previous definition, showing that convergence is faster when the invariant subspace is very close to $\mathcal{W}$, or what is the same that $\Theta(\mathcal{U} = A\mathcal{K}, A\mathcal{W}) \approx 0$.

Theorem 1. Assume that a minimal residual projection method is applied to A using the augmented Krylov subspace $\mathcal{K} = \mathcal{K}_m + \mathcal{W}$, in which the subspace $A\mathcal{W}$ is at a gap of ϵ from a certain invariant subspace $\mathcal{U}$, i.e., $\Theta(\mathcal{U}, A\mathcal{W}) = \epsilon$. Let $P_{\mathcal{U}}$ be any projector onto $\mathcal{U}$. Then, the residual $\tilde{r}$ obtained from the minimal residual projection process onto the augmented Krylov subspace $\mathcal{K}$ satisfies the inequality,

$$\|\tilde{r}\|_2 \leq \min_{q \in \mathbb{P}_m^*} \{ \|q(A)(I - P_{\mathcal{U}})r_0\|_2 + \epsilon \|q(A)P_{\mathcal{U}}r_0\|_2 \}. \quad (1)$$

Proof. By definition $\|\tilde{r}\|_2 = \min_{z \in \mathcal{K}_m + \mathcal{W}} \|r_0 - Az\|_2 = \min_{v \in \mathcal{K}_m, w \in \mathcal{W}} \|(r_0 - Av) - Aw\|_2$. Then,

$$\begin{aligned} \|\tilde{r}\|_2 &= \min_{q \in \mathbb{P}_m^*, w \in \mathcal{W}} \|q(A)r_0 - Aw\|_2 \\ &= \min_{q \in \mathbb{P}_m^*, w \in \mathcal{W}} \|q(A)(I - P_{\mathcal{U}})r_0 + q(A)P_{\mathcal{U}}r_0 - Aw\|_2 \\ &\leq \min_{q \in \mathbb{P}_m^*, w \in \mathcal{W}} \|q(A)(I - P_{\mathcal{U}})r_0\|_2 + \underbrace{\|q(A)P_{\mathcal{U}}r_0 - Aw\|_2}_{\leq \epsilon \|q(A)P_{\mathcal{U}}r_0\|_2}. \end{aligned}$$

□

This bound for $\tilde{r}$ can be exploited in multiple ways depending on the choice of the projector $P_{\mathcal{U}}$. Below, this is demonstrated for the cases when $P_{\mathcal{U}}$ is a spectral projector (deflated-Krylov procedure [5]) and when $P_{\mathcal{U}}$ is an orthogonal projector with A being a symmetric positive definite (SPD) matrix.

3.2 Deflated-Krylov procedure

Assume in Thm. 1 that $P_{\mathcal{U}}$ is the spectral projector associated to eigenvalues $\lambda_1, \dots, \lambda_s$, for $s \leq p$ and let $q_m^* \in \mathbb{P}_m^*$ be the optimal polynomial of a GMRES(m) procedure for the deflated initial system $Ax = (I - P_{\mathcal{U}})r_0 = r_d$ resulting by removing from r_0 the $s \leq p$ components associated with subspace $\mathcal{U}$, such that

$$\|q_m^*(A)r_d\|_2 = \min_{q \in \mathbb{P}_m^*} \|q(A)r_d\|_2 =: \|\tilde{r}_d\|_2.$$

Then, by Thm. 1

$$\begin{aligned} \|\tilde{r}\|_2 &\leq \|q_m^*(A)r_d\|_2 + \epsilon \|q_m^*(A)P_{\mathcal{U}}r_0\|_2 \\ &= \|\tilde{r}_d\|_2 + \epsilon \|q_m^*(A)P_{\mathcal{U}}r_0\|_2. \end{aligned}$$

If A diagonalizable, $A = UDU^{-1}$, where U is the matrix whose columns are the eigenvectors of A , and D the diagonal matrix with its eigenvalues, then $P_{\mathcal{U}}r_0$ can be expressed in the eigenbasis as $P_{\mathcal{U}}r_0 = \sum_{i \leq s} \alpha_i u_i$ which yields

$$\|\tilde{r}\|_2 \leq \|\tilde{r}_d\|_2 + \epsilon \left\| \sum_{i \leq s} q_m^*(\lambda_i) u_i \alpha_i \right\|_2.$$

Thus, for $\epsilon \approx 0$ and $\mathcal{U}$ associated with $\lambda_i \approx 0$, we have $q_m^*(0) = 1$, so we can expect the powerful result that the augmented method behaves approximately as a deflated GMRES procedure

$$\|\tilde{r}\|_2 \leq \|\tilde{r}_d\|_2. \quad (2)$$

3.3 Hermitian case

Now, let $P_{\mathcal{U}}$ be a orthogonal projector onto an invariant subspace $\mathcal{U}$ and assume there exists $q \in \mathbb{P}_m^*$ such that the components of the augmented residual in $\mathcal{K}_m$ and $\mathcal{W}$ are bounded by $s_m, c_m \in \mathbb{R}$ such that

$$\begin{aligned} \|q(A)(I - P_{\mathcal{U}})r_0\|_2 &\leq s_m \|(I - P_{\mathcal{U}})r_0\|_2 \\ \|q(A)P_{\mathcal{U}}r_0\|_2 &\leq c_m \|P_{\mathcal{U}}r_0\|_2. \end{aligned}$$

Then, plugging these expressions into (1) and since $P_{\mathcal{U}}$ is orthogonal we have

$$\begin{aligned} \|\tilde{r}\|_2 &\leq s_m \|(I - P_{\mathcal{U}})r_0\|_2 + \epsilon c_m \|P_{\mathcal{U}}r_0\|_2, \\ &= \sqrt{s_m^2 \|(I - P_{\mathcal{U}})r_0\|_2^2 + \epsilon^2 c_m^2 \|P_{\mathcal{U}}r_0\|_2^2} \end{aligned}$$

and after applying the Cauchy-Schwarz inequality the inequality yields

$$\|\tilde{r}\|_2 \leq \sqrt{s_m^2 + \epsilon^2 c_m^2} \|r_0\|_2. \quad (3)$$

If we also assume that the invariant subspace $\mathcal{U}$ is the subspace associated with $\lambda_1, \dots, \lambda_s$ for $s \leq p$, and in the particular case A is SPD with $0 < \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$,

then by using approximation theory results based on Chebyshev polynomials, upper bounds for $s_m \leq \frac{1}{T_m(\gamma)}$ and $c_m \leq 1$ are found in [1]:

$$\|\tilde{r}\|_2 \leq \sqrt{s_m^2 + \epsilon^2 c_m^2} \|r_0\|_2 \leq \sqrt{\frac{1}{T_m^2(\gamma)} + \epsilon^2} \|r_0\|_2 \quad (4)$$

in which $\gamma = \frac{\lambda_n + \lambda_{s+1}}{\lambda_n - \lambda_{s+1}}$ and T_m is the Chebyshev polynomial of degree m of the first kind.

The above result aligns with the previously observed outcome for the general deflated-Krylov case, where for $\epsilon \approx 0$, the residual $\tilde{r}$ approaches the residual of the system without the smallest eigenvalues. The ratio $\gamma(\lambda_n, \lambda_{s+1})$ it is somehow related with the condition number of the deflated system matrix, $\kappa = \frac{\lambda_n}{\lambda_{s+1}}$. It is also worth noting in this context that very close values for λ between $s+1$ and n (clusters of eigenvalues) will produce high values of $T_m(\gamma)$, or in other words, low values for the bound of $\tilde{r}$, consistent with the convergence result of the GMRES method for normal or approximate normal matrices [6]

$$\frac{\|r_m\|_2}{\|r_0\|_2} \leq \kappa(V) \inf_{p \in \mathbb{P}_m^*} \max_{\lambda \in \sigma(A)} |p(\lambda)|,$$

where $\sigma(A)$ is the spectrum of A and V the matrix of its spectral decomposition. For highly non-normal matrices, this result is not applicable since the condition number of the eigenvector matrix can be very high, causing this bound for $\tilde{r}$ to blow up.

4 Block-Krylov methods

Now, we deal with a slightly different kind of problem. In many fields, problems of the type $AX = B$ arise where we have the same system matrix A but multiple right-hand sides. Here, $X, B \in \mathbb{R}^{n \times s}$ are block vectors representing multiple solutions and corresponding right-hand sides, respectively. In this case, $\mathcal{W}$ consists of the sum of a few other standard Krylov subspaces generated by

$$\mathcal{K} = K_m^{(1)} + \underbrace{K_m^{(2)} + K_m^{(3)} + \dots + K_m^{(s)}}_{\mathcal{W}}$$

where

$$\mathcal{K}_m^{(i)}(A, v_1^{(i)}) = \text{span} \left\{ v_1^{(i)}, Av_1^{(i)}, \dots, A^{m-1}v_1^{(i)} \right\}$$

with normalized initial residual $v_1^{(i)} = \frac{r_0^{(i)}}{\|r_0^{(i)}\|_2}$.

4.1 General result

The subspace where we are going to project the residuals in this case is the subspace spanned by the initial block $\mathcal{S} := \text{span} \left\{ v_1^{(1)}, v_1^{(2)}, \dots, v_1^{(s)} \right\}$.

Given any subspace $\mathcal{U}$ of $\dim = s$, a projector is uniquely defined by two subspaces:

$$\text{Range}(P_{\mathcal{U}}) = M = AS, \quad \text{Null}(P_{\mathcal{U}}) = N = \mathcal{L}^{\perp}$$

where $P_{\mathcal{U}}$ is a projector onto $\mathcal{U}$ and orthogonal to $\mathcal{L}$ with condition

$$AS \cap \mathcal{L}^{\perp} = \{\mathbf{0}\} \Leftrightarrow AS \oplus \mathcal{L}^{\perp} = \mathbb{C}^n.$$

If we consider $w \in \mathcal{S}$ such that Aw is the (unique) orthogonal projection w.r.t. $\mathcal{L}$ of some $r \in \mathbb{C}^n$ onto AS , then by projection theory

$$\underbrace{r - Aw}_{\hat{r}} \perp \mathcal{L} \implies P_{\mathcal{U}}(r - Aw) = 0 \Leftrightarrow r - Aw \in \text{Null}(P_{\mathcal{U}}) = \mathcal{L}^{\perp}.$$

Combining all these results, the following bound for $\tilde{r}$ is derived:

Theorem 2. Let $P_{\mathcal{U}}$ be a projector onto a subspace $\mathcal{U}$ of dimension $\dim = s$, such that $AS \cap \mathcal{L}^{\perp} = \{\mathbf{0}\}$, i.e., $\mathbb{C}^n = AS \oplus \mathcal{L}^{\perp}$ is satisfied for $\mathcal{L} = \text{Null}(P_{\mathcal{U}})^{\perp}$. Let w_0 be the vector that satisfies $P_{\mathcal{U}}(r_0 - Aw_0) = 0$ and denote $\hat{r}_0 = r_0 - Aw_0$, then,

$$\|\tilde{r}\|_2 \leq \min_{q \in \mathbb{P}_m^*} \{ \|q(A)(I - P_{\mathcal{U}})\hat{r}_0\|_2 \}. \quad (5)$$

Proof. By definition $\|\tilde{r}\|_2 = \min_{z \in \mathcal{K} = \mathcal{K}_m + \mathcal{W}} \|r_0 - Az\|_2 = \min_{v \in \mathcal{K}_m, w \in \mathcal{K}} \|(r_0 - Av) - Aw\|_2$. Then,

$$\begin{aligned} \|\tilde{r}\|_2 &= \min_{q \in \mathbb{P}_m^*, w \in \mathcal{K}} \|q(A)r_0 - Aw\|_2 \\ &= \min_{q \in \mathbb{P}_m^*, w \in \mathcal{K}} \|q(A)(I - P_{\mathcal{U}})r_0 + q(A)P_{\mathcal{U}}r_0 - Aw\|_2 \\ &\leq \|q(A)(I - P_{\mathcal{U}})r_0 + q(A)P_{\mathcal{U}}r_0 - Aw\|_2. \end{aligned}$$

$$\begin{aligned} q(A)P_{\mathcal{U}}r_0 - Aw &= q(A)P_{\mathcal{U}}r_0 - Aq(A)w_0 = q(A)P_{\mathcal{U}}r_0 - q(A)Aw_0 \\ [\text{from cond. } P_{\mathcal{U}}(r_0 - Aw_0) = 0 \Leftrightarrow Aw_0 &= P_{\mathcal{U}}r_0 + (I - P_{\mathcal{U}})Aw_0] \\ &= q(A)P_{\mathcal{U}}r_0 - q(A)[P_{\mathcal{U}}r_0 + (I - P_{\mathcal{U}})Aw_0] \\ &= -q(A)(I - P_{\mathcal{U}})Aw_0 \end{aligned}$$

Thus,

$$\|\tilde{r}\|_2 \leq \|q(A)(I - P_{\mathcal{U}})(r_0 - Aw_0)\|_2.$$

Taking the $\min_{q \in \mathbb{P}_m^*} q(A)$ gives the desired result. $\square$

Residual bound from (5) suggests that a block-GMRES will approximately behave like a GMRES with residual $\hat{r}_0$, which is the residual obtained by removing from r_0 components in $\mathcal{U}$ through projection onto the initial block $\mathcal{S}$. This removal process is achieved by projecting the residuals onto AS orthogonally to $\mathcal{L} = \text{Null}(P_{\mathcal{U}})^{\perp}$ satisfying the Galerkin conditions [7]

$$w_0 \in \mathcal{S}, \quad r_0 - Aw_0 \perp \mathcal{L}.$$

4.2 Hermitian case

As in deflated-Krylov case, if we assume A is SPD with eigenvalues

$$0 < \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$$

and $P_{\mathcal{U}}$ the orthogonal projector onto invariant subspace $\mathcal{U}$ associated with eigenvalues $\lambda_1, \dots, \lambda_p$. Then, assuming conditions of Thm. 2, the result for the augmented residual using Chebyshev polynomials yields

$$\|\tilde{r}\|_2 \leq \frac{\|\hat{r}_0\|_2}{T_m(\gamma)} \quad (6)$$

with $\gamma \equiv \frac{\lambda_n + \lambda_{p+1}}{\lambda_n - \lambda_{p+1}}$.

Once again, similar to the deflated-Krylov case, a clear dependence of the augmented residual value on the condition number of the deflated system matrix is observed, and ultimately on the eigenvalue distribution in the spectrum of such matrix.

5 Numerical experiment

The numerical experiment in Saad's article [1] builds a sparse linear system with a simple diagonal matrix A of size $n = 200$, entrywise defined by

$$d_i = \begin{cases} \frac{i}{n} & \text{when } i > 4 \\ 0.05 \times \frac{i}{n} & \text{when } i \leq 4, \end{cases}$$

and b randomly chosen.

The spectrum of this matrix

$$\sigma(A) = \{0.00025, 0.0005, 0.00075, 0.001, 0.025, 0.03, \dots, 1\}$$

is specially chosen to have an initial cluster around zero of size 4 and an equally spaced eigenvalue distribution from entries 5 to 200. These linearly spaced eigenvalues between 0.025 and 1 do not form tight clusters, but the spacing is regular and not too wide. This could still be manageable for GMRES, but it might not be as beneficial as having more tight clusters. This matrix structure is expected to show an improvement in convergence of residual norms with respect to standard Krylov methods to support bound results obtained for $\|\tilde{r}\|_2$, that is, that the augmented residual will approximate the calculation of the deflated system's residual.

To that aim, the following five experiments were run with the same RHS $b = \text{randn}(n, 4)$, $x_0 = 0$, $\text{tol} = 10^{-10}$, and $\text{maxiter} = 180$:

- Full standard and restarted GMRES (Krylov dim. $m = 40$), denoted by GMRES(inf) and GMRES(40) respectively.
- Block-GMRES without restarts, block size = 4 (four linear systems simultaneously) denoted by BGMRES(inf,4).
- Deflated-GMRES augmented with approximate eigenvectors for the first 4 eigenvalues which is denoted by DGMRES(40, 4).
- For comparison, a GMRES without restart on the deflated system by removing the four first entries of the system, i.e., $d_i, b_i, 5 \leq i \leq 200$, denoted by GMRES(deflated sys.).

Results of the experiment are shown in Figures 1 and 2.

Figure 1 is implemented by using the FGMRES [3] Algorithm 1 seen in this report. The test uses an augmented Krylov subspace of dimension 40, the last four of which are the eigenvectors corresponding to the four smallest eigenvalues. This explains why just before iteration 40, when these eigenvectors come into play, we can see how the convergence for the residual norm is improved to almost reach the deflated system ratio.

Figure 2 shows the behaviour of both block- and deflated-Krylov, but in this case the standard Krylov subspace is augmented with eigenvectors computed from previous Arnoldi steps (Ritz vectors) and that is why the convergence for DGMRES needs an outer loop more. Then, it is around iteration 80 when it is observed a clear acceleration of convergence. Full GMRES or GMRES(inf) outperforms other methods, possibly due to using a subspace containing the same eigenvectors as DGMRES(40, 4) after iteration 40. It is also appreciated that BGMRES tends to take longer to capture clusters and reach final convergence but at the end it averages 45 steps per system. All these experimental results are consistent with the initial predicted expectations, confirming the theoretical outcomes obtained in the article.

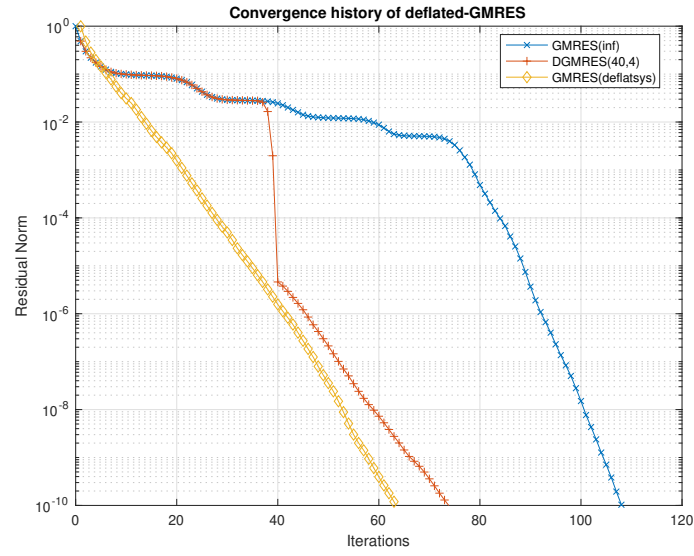


Figure 1: Deflated-GMRES procedure.

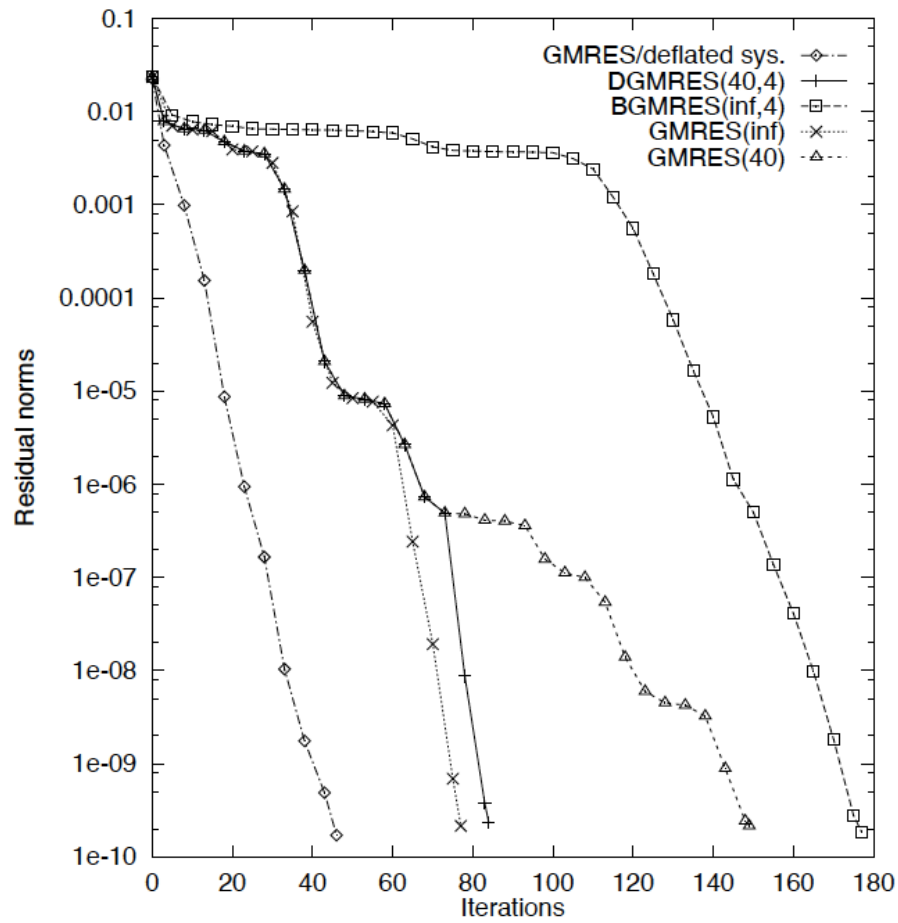


Figure 2: Deflated and block-GMRES behaviour analysis [1].

6 Conclusion

Based on the analysis and the results obtained from the previous numerical experiment, the following conclusions can be drawn:

- (i) All methods except restarted GMRES exhibit similar convergence slopes towards the final phase of iteration, meaning that the convergence ratio for the augmented techniques are similar to the deflated system one.
- (ii) Block-GMRES generally requires more time to identify clusters and achieve complete convergence.
- (iii) Instead, for solving simultaneous linear systems, block-GMRES could be very competitive.
- (iv) For a single linear system, full or deflated-GMRES may achieve superior performance, though the latter will be more suitable for large linear systems due to its lower memory demand.

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